

MANIKANDAN MUTHIAH

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LinkedIn: [linkedin.com/in/manikandan-muthiah-6a096214b](https://www.linkedin.com/in/manikandan-muthiah-6a096214b) Portfolio: pcbzmani.github.io/portfolio

PROFESSIONAL OBJECTIVE

To be a dynamic professional with a passion for learning new technologies, adapting to evolving environments, and delivering best-in-class data engineering solutions.

TECHNICAL SUMMARY

- Senior data engineer with expertise in PySpark, Python, SQL and strong exposure to Azure and GCP cloud.
- Around 9.6 years of software industry experience, extensively in development projects with multiple deployments, Go-Live implementation and upgrades.
- Around 8.6 years of experience in data engineering.
- Around 7.6 years of experience in Agile workflow model.
- Around 8.6 years of experience in Azure cloud services like ADF, Azure Databricks.
- Around 2.6 years of experience in GCP cloud services like Dataproc, Composer and BigQuery.
- Extensive knowledge in Healthcare, Mortgage Insurance, Retail and Airline domains.
- Bachelor of Engineering in Electrical and Electronics, National Engineering College with 91%.

EDUCATION

- Bachelor of Engineering in Electrical and Electronics, National Engineering College with 91%.

CERTIFICATION

- Certified in Azure Data Engineer (DP-203)
- Certified in Databricks – Data Engineer Associate and Data Analyst
- Certified in Google Professional Data Engineer Associate

JOB PROFILE

- Senior Software Engineer at EMAKINA (EPAM Systems), Doha – Sep 2023 – Till date
- Senior Software Engineer at EPAM Systems, Chennai – Apr 2022 – Sep 2023
- Application Development Senior Analyst at Accenture, Chennai – Apr 2021 – April 2022
- IT Analyst at TCS, Chennai – Jul 2016 – Apr 2021

TECHNICAL SKILLS

Programming Languages	Python, SQL
Cloud Platforms	Azure, GCP
Azure & GCP Tools	Azure Data Factory, Azure Databricks, Dataproc, Composer (Airflow), BigQuery
Database	MS SQL, Oracle, SQL Server, PostgreSQL, Delta Tables
AI / LLM	Large Language Models (LLMs), RAG (Retrieval-Augmented Generation), MCP Servers

PROJECT DETAILS

Project: Duty Free Business Intelligence

Client: Leading Aviation in Qatar

Project Description:

A global aviation duty-free service company that offers tax-exempt shopping for luxury goods, travel essentials, and exclusive travel retail experiences, along with loyalty programs, primarily to international travellers at airports and key transit hubs worldwide.

Role: Senior Data Engineer

Environment: Azure Databricks, Google Cloud Platform (GCP), GCP BigQuery, Google Composer (Airflow), PowerBI

Duration: Sep 2023 – Present

Role & Responsibilities:

- Engaged in requirement analysis, design, and development of a scalable ingestion framework to transfer data from on-premise systems to Google Cloud.
- Transformed raw data to generate actionable insights, enabling the data science team to forecast duty-free sales trends and optimize food waste management for in-flight services.
- Prepared and fed curated datasets into PowerBI, empowering business stakeholders to track sales performance and make data-driven decisions.
- Orchestrated end-to-end data pipelines using Google Composer (Airflow) for seamless workflow management and scheduling.
- Optimized long-running jobs by implementing best practices such as indexing in BigQuery and efficient partitioning, reducing processing time significantly.
- Developed LLM-powered features and Retrieval-Augmented Generation (RAG) pipelines to enable intelligent querying and insight generation from aviation and retail datasets.
- Built and maintained Model Context Protocol (MCP) servers to support structured, context-aware interactions between AI agents and internal data sources.

Project: Aviation Inventory Analytics

Client: *Leading Aviation in Qatar*

Project Description:

A data and analytics initiative focused on aviation inventory management, providing insights into booking predictions, transaction patterns, and comprehensive flight operations (Flight 360) to enhance operational efficiency.

Role: Senior Data Engineer

Environment: Azure Databricks, Delta Tables, Dremio

Duration: Jul 2022 – Sep 2023

Role & Responsibilities:

- Designed and developed an ingestion framework to load data from on-premise systems to Azure cloud, ensuring high availability and scalability.
- Transformed complex datasets to extract insights, supporting the data science team in forecasting booking trends, transaction behaviours, and Flight 360 metrics.
- Enhanced performance of long-running jobs by applying best practices such as Delta table optimizations and query tuning.
- Extensively used joins and sub-queries to integrate multi-source data for comprehensive analytics.
- Documented processes and shared technical insights to improve team collaboration.

Project: Data Factory for CTC

Client: *Retail & Distribution*

Project Description:

A data warehousing solution to centralize and process retail and distribution data, enabling efficient inventory management and operational insights through cloud-based analytics.

Role: Senior Data Engineer

Environment: Azure Data Factory, Azure Databricks, Delta Lake, Azure MySQL, MSSQL, Bitbucket, Jenkins, qTest, JIRA, Confluence, HDFS

Duration: Apr 2022 – Jun 2022

Role & Responsibilities:

- Developed a file- and RDBMS-based ingestion framework to transfer data from on-premise systems to Azure cloud with minimal latency.
- Created and optimized pipelines in Azure Data Factory to extract data from on-premise sources and load it into Azure Databricks Delta Lake.
- Improved Delta Lake performance by compacting small files into larger ones, reducing query execution time.
- Applied Spark optimizations including broadcast joins, repartitioning, coalesce, and salting to enhance processing efficiency.
- Documented workflows in Confluence and tracked progress via JIRA for effective project management.

Project: Home Buyer Transformation

Client: *NA Mortgage Insurance*

Project Description:

A program leveraging big data and AI to assist homebuyers with mortgage insurance, enabling rapid and informed business decisions through advanced analytics.

Role: Data Engineer

Environment: Azure Data Factory, Azure Databricks, Azure SQL, PySpark, Hive, Azure DevOps, VS Code

Duration: Apr 2021 – Apr 2022

Role & Responsibilities:

- Delivered code iteratively across sprints, ensuring timely feature deployment.
- Utilized Azure Data Factory (ADF) to ingest data from diverse sources into Azure Data Lake for centralized storage.
- Designed and implemented data pipelines in ADF, integrating Databricks for transformation and loading data into target databases.
- Leveraged PySpark for data transformation, applying best practices and Spark optimizations (e.g., caching, partitioning) to enhance performance.

Project: Healthcare Data Modernization

Client: *US Healthcare*

Project Description:

A global health insurance company offering health, dental, vision, life, and accidental insurance along with Medicare and Medicaid products, modernizing legacy systems for enhanced data processing and reporting.

Role: Developer

Environment: MS SQL, IBM DB2, Sqoop, Hive, PySpark, HDFS, AWS Redshift, Python, Spark

Duration: Jul 2016 – Apr 2021

Role & Responsibilities:

- Converted legacy COBOL SQL code into optimized Hive queries, improving query performance significantly.
- Created Hive tables to view and analyse data stored in HDFS, incorporating optimization techniques like bucketing and indexing.
- Implemented PySpark with DataFrames API and Spark SQL for faster processing, applying optimization techniques such as broadcast joins and repartitioning.

I do hereby declare that the information furnished above is true to the best of my knowledge and belief.

Yours faithfully,

Place: Doha, Qatar

Manikandan Muthiah